Sanjay Krishna

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Summary

Hardware engineer who owns designs end to end, from micro-architecture and RTL to verification, PnR and signoff, judging every decision by power, performance and area. Taped out a Tiny Tapeout SNN accelerator (sub-10 nW idle) and took a 16×16 mixed-precision NPU through clean GDSII signoff. Motivated by making compute more efficient. Looking for internships with Intent to return to degree-program after the completion of the internship

Projects

16×16 Mixed-Precision Systolic NPU for Edge-AI Inference | GitHub Jan 2026 – Oct 2026

- Architected a 16×16 mixed-precision systolic NPU in synthesizable Verilog with reconfigurable sub-bit fusion (INT8/INT4/INT2), an on-chip Adaptive Precision Controller (APC) for runtime throttling, and Zero-Activation Gating (ZAG) to eliminate redundant switching power.
- Implemented the full ASIC RTL - GDSII flow to clean GDSII signoff in Cadence Suite Using TCL Scripts on 45nm library **1.23 mm^2 core area**, **135.8 mW total power** at 166 MHz (0.085–0.341 TOPS, 0.63–2.51 TOPS/W), with macro floorplanning, CTS and ECO pin-access resolution.
- Built a bit-exact RTL-to-Python co-simulation harness validating end-to-end CIFAR-10 edge inference; held >**91% accuracy** under dynamic runtime precision scaling on ResNet-20 and VGG-16 with up to 34% dynamic energy reduction.

Event-Driven Spiking Neural Network ASIC for AFib Detection | GitHub Feb 2026 – April 2026

- Closed timing at **0 ns worst-negative-slack** and passed **DRC/LVS/antenna signoff clean** within **959/1000** standard cells (**0.012 mm^2** core) for a processor-less SNN accelerator **selected for fabrication on the Tiny Tapeout SKY26B shuttle**, owning RTL through GDSII signoff in OpenLane using python scripts.
- Achieved a **sub-10 nW** idle power floor (vs. 1.884 mW peak) via a 5-stage event-driven pipeline and area-conscious PPA techniques (multiplier-free arithmetic, reduced-width neuron state) that fit the 1000-cell budget and keep the datapath idle between heartbeats.
- Validated on **15,924 windows** of real clinical ECG data (MIT-BIH AFib DB) via RTL hardware-in-the-loop testing, with a patient-level held-out split to avoid the data leakage common in embedded-classifier evaluation.

PUF-Based Hardware Security Module on FPGA Jul 2026 – Aug 2026

- Designed a lightweight, tamper-resistant FPGA Hardware Security Module (HSM) with a 32-node Ring Oscillator PUF (RO-PUF) and BCH fuzzy extractor that reconstructs a 128-bit cryptographic key on the fly.
- Integrated an ultra-compact PRESENT-128 block cipher with volatile, on-the-fly key generation, removing non-volatile key storage as a target for physical probing and side-channel extraction.
- Mitigated systematic routing and placement bias across the FPGA fabric with custom Vivado TCL floorplanning constraints enforcing symmetric, differential pairing of co-located oscillator slices for optimal inter-die uniqueness.

Experience

Research Intern, CSIR-CEERI, Chennai May 2026 – June 2026

- Accelerated real-time NIR spectroscopic classification by >40× over PC software, to a deterministic 20.94 μs latency (47.6k spectra/s), using pipelined Verilog **Digital Signal Processing** datapaths for Savitzky–Golay FIR filtering, baseline subtraction and SNV normalization.
- Closed 100 MHz timing on a Xilinx Artix-7 (XC7A100T) in Vivado, mapping parallel fixed-point SVM inference onto DSP48E1 slices with a single-cycle combinational Argmax tree for 92.5% 5-class sorting accuracy.
- Owned FPGA subsystem interfaces in a hardware-software co-design framework: a 115200-baud UART streaming bridge to the host and an autonomous BRAM/VIO testbench for in-silicon verification.

Research Intern, Department of ECE, NIT Trichy May 2025 – June 2025

- Delivered a direct **PPA** gain over exact multiplication with a prioritized Sobel edge-detector architecture built on compressor-based approximate multipliers, trading bounded precision loss for reduced multiplier area.
- Closed functional **verification** on two independent IC designs (edge-detector, 8-bit CPU datapath) with self-checking FPGA testbenches and MATLAB golden-model comparison, driving verification through simulation, waveform analysis and debug.

Skills

Micro-architecture & RTL: Verilog, VHDL, Lint (Verilator), RTL Design and Synthesis, Micro-architecture Development, Computer Architecture, Datapath Development, CDC

PPA, Verification & Physical Design: Static Timing Analysis (STA), Power/Performance/Area Optimization, Coverage-Driven Functional Verification (directed, constrained-random), Physical Design (PnR, DRC/LVS), ASIC Flow, FPGA System Design

EDA Tools: Cadence Suite, OpenLane, Modelsim, Quartus, Xilinx Vivado

Programming & Scripting: Python, TCL, C, Linux

Education

Vellore Institute of Technology, Vellore – B.Tech in Electronics and Communication Engineering 2023 – 2027

- CGPA: 9.29/10
- Coursework: Digital Logic Design, Microcontrollers, VLSI System Design, Computer Architecture, RTL-to-GDS2 Flow